SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I Beffy A Shanika(192511387) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

beffy

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

Selected Article

Title: Reinforcement learning assisted oxygen therapy for COVID-19 patients under intensive care

Authors: Hua Zheng, Jiahao Zhu, Wei Xie, et al.

Journal: BMC Medical Informatics and Decision Making

Year: 2021

Volume: 21, Article 350

DOI: https://link.springer.com/article/10.1186/s12911-021-01712-6

The article develops a deep Reinforcement Learning system that recommends personalized oxygen-flow rates for critically ill COVID-19 patients.

TASK 1: ARTICLE SUMMARY

(a) Citation, Domain and ML Problem

Full Citation

Zheng, H., Zhu, J., Xie, W., et al. (2021). "Reinforcement learning assisted oxygen therapy for COVID-19 patients under intensive care." BMC Medical Informatics and Decision Making, 21, Article 350. DOI: 10.1186/s12911-021-01712-6.

Domain

The article belongs to the healthcare and intensive-care domain.

Specific ML Problem

The problem is to determine an appropriate oxygen flow rate continuously for critically ill COVID-19 patients.

The researchers formulate oxygen therapy as a Markov Decision Process (MDP) and use Deep Deterministic Policy Gradient (DDPG) to learn a personalized oxygen-treatment policy.

(b) Article Summary – 150–200 Words

The article proposes a Reinforcement Learning based method to provide personalized oxygen therapy for critically ill COVID-19 patients. The main objective is to determine an appropriate oxygen flow rate according to the changing health condition of each patient. The authors identified that oxygen therapy is an important treatment for COVID-19 patients with respiratory problems, but selecting the correct flow rate continuously is difficult because patient conditions change over time. To address this problem, the researchers modeled patient health trajectories as a Markov Decision Process. Patient demographic information, vital signs and laboratory test results were used as the state, while oxygen flow rate from 0 to 60 L/min was considered the action. The reward was based on the patient's 7-day survival outcome. A Deep Deterministic Policy Gradient algorithm was then used to learn an optimal treatment policy. The study used retrospective electronic health records from NYU Langone Health and evaluated the model using cross-validation. The results showed that the RL-based approach estimated lower mortality than standard physician-prescribed oxygen therapy and could potentially support personalized clinical decision-making.

TASK 2: METHODOLOGY ANALYSIS

(a) ML Technique and Dataset

ML Technique

The researchers used Deep Deterministic Policy Gradient (DDPG).

DDPG is a deep Reinforcement Learning algorithm suitable for continuous action spaces. In this study, oxygen flow is a continuous action ranging from 0 to 60 L/min.

The DDPG architecture contains:

- Actor network – selects the oxygen-flow action.
- Critic network – evaluates the selected action using the Q-value.

- Target networks – help stabilize training.
- Experience replay – stores historical patient transitions for learning.

The actor produces an oxygen flow rate for a given patient state, while the critic estimates the quality of that action.

Dataset

The study used retrospective Electronic Health Record (EHR) data from New York University Langone Health (NYULH).

The article reports 1,372 critically ill COVID-19 patients in its abstract; the detailed results section reports 1,362 patients in the analysis cohort. This discrepancy should be acknowledged rather than silently corrected.

The dataset included:

- Age
- Sex
- Race
- Smoking status
- BMI
- Comorbidities
- Vital signs
- Laboratory test results
- ICU information
- Treatment information
- Patient outcomes

The researchers selected 36 laboratory tests, using criteria including less than 28% missing values and relevance to COVID-19 tests/vital signs.

Data Preparation

The data was:

1. Collected from EHR records.
2. De-identified for privacy.
3. Filtered according to patient eligibility.
4. Reduced to relevant laboratory and clinical features.
5. Divided using leave-one-hospital-out validation.

The whole dataset was divided into four hospital-based batches; three were used for training and one for validation in each simulation.

(b) Mapping to CO4

CO4 Topic	Application in Article
Inductive Learning	The system learns treatment policies from historical patient data
Decision Trees	Not used as the main algorithm
Statistical Learning	Cox proportional hazards model was used to estimate survival outcomes
Reinforcement Learning	DDPG learns an optimal oxygen-treatment policy

The strongest connection is to Reinforcement Learning, because the system learns actions based on patient states and rewards.

Methodological Strength

A major strength is that the method handles continuous treatment decisions. Oxygen flow can vary continuously between 0 and 60 L/min, making DDPG appropriate for the problem. The MDP formulation also considers delayed health outcomes rather than only immediate results.

Limitation

One limitation is that the study uses retrospective observational data, so the actual outcomes of some RL-recommended treatments were not directly observed. The researchers therefore estimated these outcomes using a Cox survival model.

TASK 3: RESULTS & CRITICAL EVALUATION

(a) Key Results

The major results reported by the authors are:

Metric	Physician/Standard Care	RL-Oxygen
Estimated 7-day mortality	7.94%	5.37%
Difference	—	2.57% lower
Average oxygen flow	—	1.28 L/min lower

The mortality reduction had a 95% confidence interval of 2.08–3.06 percentage points and $P < 0.001$.

The survival prediction model also reported:

- Cosine similarity: >99.9%
- Concordance index: 0.83
- Paired Chi-squared test: $p < 0.0001$

The RL system was considered consistent with physicians when the recommended and observed oxygen flow rates differed by less than 10 L/min. The authors reported that the rates were within 10 L/min about 44% of the time.

(b) Critical Evaluation

Are the Results Realistic?

The results are promising because the RL system estimated a reduction in mortality from 7.94% to 5.37%. However, these results should not be interpreted as proof that the RL system would definitely reduce mortality in real clinical practice.

The study itself acknowledges that the evaluation is based on retrospective data and that randomized clinical trials would ultimately be needed to validate the treatment policy.

Potential Biases

1. Dataset Bias

The data came from NYU Langone Health, so it may not represent all COVID-19 patients or hospitals.

2. Missing Data

The authors state that sample scarcity and missing values may increase uncertainty in the treatment recommendations.

3. Geographic Bias

Clinical practices in other countries may be different from those represented in the NYULH dataset.

4. Retrospective Data Bias

The model learns from treatments that were already given by doctors. Therefore, it may inherit patterns or biases present in historical clinical decisions.

Additional Experiments

The study could be strengthened by:

- Testing the model on datasets from multiple countries.
- Conducting prospective clinical trials.
- Comparing DDPG with other RL algorithms.
- Testing the system on a larger patient population.
- Performing subgroup fairness analysis.
- Testing robustness under different missing-data conditions.

The authors themselves state that larger training data and randomized clinical trials would be valuable for validating the method.

(c) Real-World Applicability

The system could potentially be deployed as a clinical decision-support tool in ICUs.

Possible applications include:

- Personalized oxygen therapy.
- ICU treatment support.
- Continuous patient monitoring.
- Resource optimization during oxygen shortages.
- Supporting doctors in treatment decisions.

The article concludes that the algorithm has potential to be integrated into clinical decision-support systems to assist physicians with timely personalized oxygen recommendations.

Challenges in Industry

Data availability: Large, high-quality EHR datasets are required.

Computational requirements: Deep RL requires significant computational resources.

Privacy: Patient health records must be protected.

Ethical concerns: Incorrect treatment recommendations can directly affect patient safety.

Explainability: Doctors need to understand and trust the AI's recommendations.

Clinical validation: The system must be carefully tested before real-world deployment.

TASK 4: REFLECTION & LEARNING OUTCOME

(a) Three Key Insights

Insight 1 – MDP in Healthcare

I learned that a healthcare treatment problem can be represented as a Markov Decision Process, where the patient's health condition is the state, treatment is the action, and patient outcome is the reward.

CO4 Connection: Reinforcement Learning and MDP.

Insight 2 – Delayed Rewards

The article showed that a treatment decision may not produce an immediate result. The final patient outcome can be used as a reward and propagated backward to earlier decisions.

CO4 Connection: Reward function and long-term decision-making in Reinforcement Learning.

Insight 3 – Continuous Actions

I learned that some real-world problems have continuous actions rather than fixed choices. In this article, oxygen flow can take values from 0 to 60 L/min, which makes DDPG suitable.

CO4 Connection: Reinforcement Learning algorithms for continuous action spaces.

(b) Proposed Extension

I would extend the research by developing a multi-hospital and multi-country RL model.

Proposed Experiment

The model can be trained using EHR data from multiple hospitals and tested on a completely new hospital.

Hospital A ──┐

Hospital B ──┼──> RL Training ──> New Hospital Testing

Hospital C ──┘

Justification

The original research recognizes that a single hospital's COVID-19 cohort may not represent patients or clinical practices elsewhere. Testing across multiple hospitals would improve generalisation and reduce dataset bias.

Expected Impact

This enhancement could:

- Improve model generalisation.
- Reduce hospital-specific bias.
- Make the system more reliable.
- Support deployment in different healthcare environments.
- Provide stronger evidence for real-world clinical use.

CONCLUSION

The reviewed article demonstrates how Reinforcement Learning can support personalized healthcare decision-making. The researchers used DDPG with EHR data to learn oxygen-flow policies for critically ill COVID-19 patients. The RL approach estimated a reduction in 7-day mortality compared with standard physician-prescribed oxygen therapy, while also recommending lower average oxygen flow.

The article provides a strong practical example of CO4 Reinforcement Learning, especially the concepts of MDP, states, actions, rewards, Q-functions, policies and continuous action spaces. However, because the study relies on retrospective data, further validation through larger datasets and randomized clinical trials is necessary before autonomous clinical deployment

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand